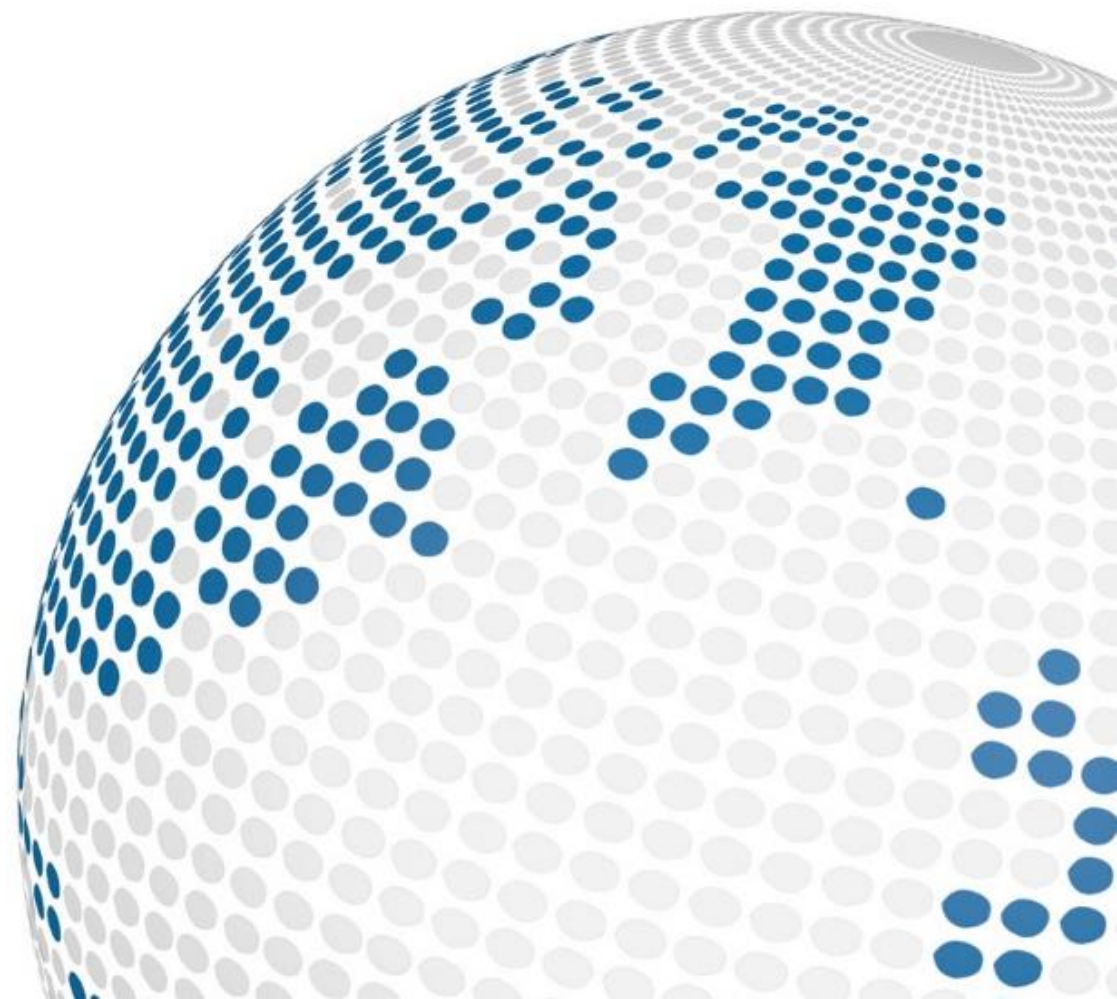


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Shaping the world with sensor solutions

July, 2022



AS7056/57 Design-in Guidelines

Overview

System Overview

Power supply Scheme

IO Signals

Proposal for Optical Design

Optical Simulation

AS7056 System Overview

Block Diagram

The AS7056 Biosignal Sensor AFE contains two main blocks. An analog front-end for LED driving, signal acquisition, photodiode selection, and signal preconditioning. Moreover, a digital backend for signal filtering, balancing, and sampling. Furthermore, the digital block will handle the sensor configuration, control, and communication to the external MCU.

AS7056 AFE (small dimension and high performance):

- PPG acquisition channels
- 2 independent acquisition channels (2 ADC)
- 3 Photodiodes
- 1 **VCSEL** driver and 2 LED drivers for external VCSELs/LEDs
- Proximity
- Ultra short conversion time
- HRM/HRV, SpO2, Blood pressure

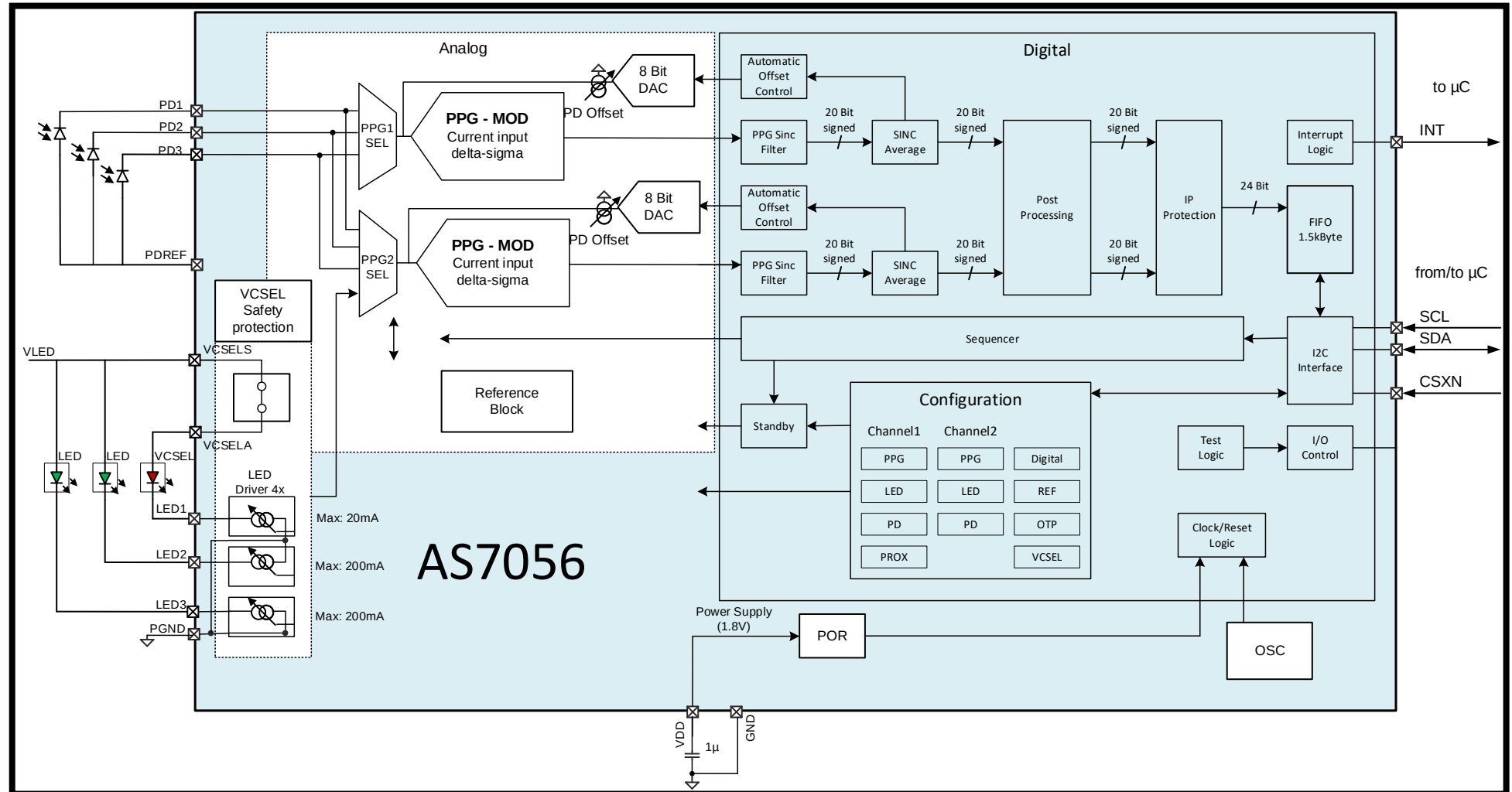


Fig. 1: AS7056 Block Diagram

AS7056 System Overview

Pin Diagram

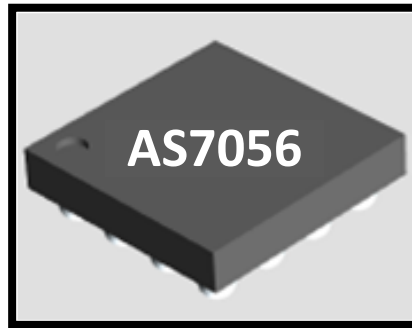


Fig 2: Top View of AS7056

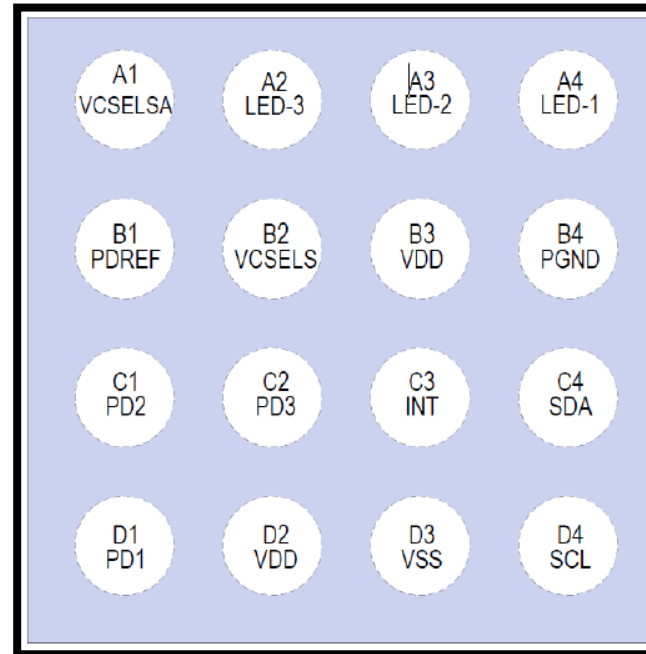


Fig. 3: AS7056/57 Pin Diagram

Pin Number	Pin Name	Description
A1	VCSELSA	VCSEL output (anode)
A2	LED3	Output LED driver 3
A3	LED2	Output LED driver 2
A4	LED1	Output LED driver 1
B1	PDREF	Reference potential for photodiodes
B2	VCSELS	VCSEL input (supply)
B3	VDD	Power Supply (typ. 1V8)
B4	PGND	Ground for LED drivers 1-3
C1	PD2	Photodiode input
C2	PD3	Photodiode input
C3	INT	Interrupt
C4	SDA	I ² C Data
D1	PD1	Photodiode input
D2	VDD	Power Supply (typ. 1V8)
D3	VSS	Ground
D4	SCL	Clock for I ² C

Fig. 4: Pin description

AS7057 System Overview

Block Diagram

The AS7057 Biosignal Sensor AFE contains two main blocks. An analog front-end for LED driving, signal acquisition, photodiode selection, and signal preconditioning. Moreover, a digital backend for signal filtering, balancing, and sampling. Furthermore, the digital block will handle the sensor configuration, control, and communication to the external MCU.

AS7057 AFE (small dimension and high performance):

- PPG acquisition channels
- 2 independent acquisition channels (2 ADC)
- 3 Photodiodes
- 3 drivers for external LEDs
- Proximity
- Ultra short conversion time
- HRM/HRV, SpO2, Blood pressure

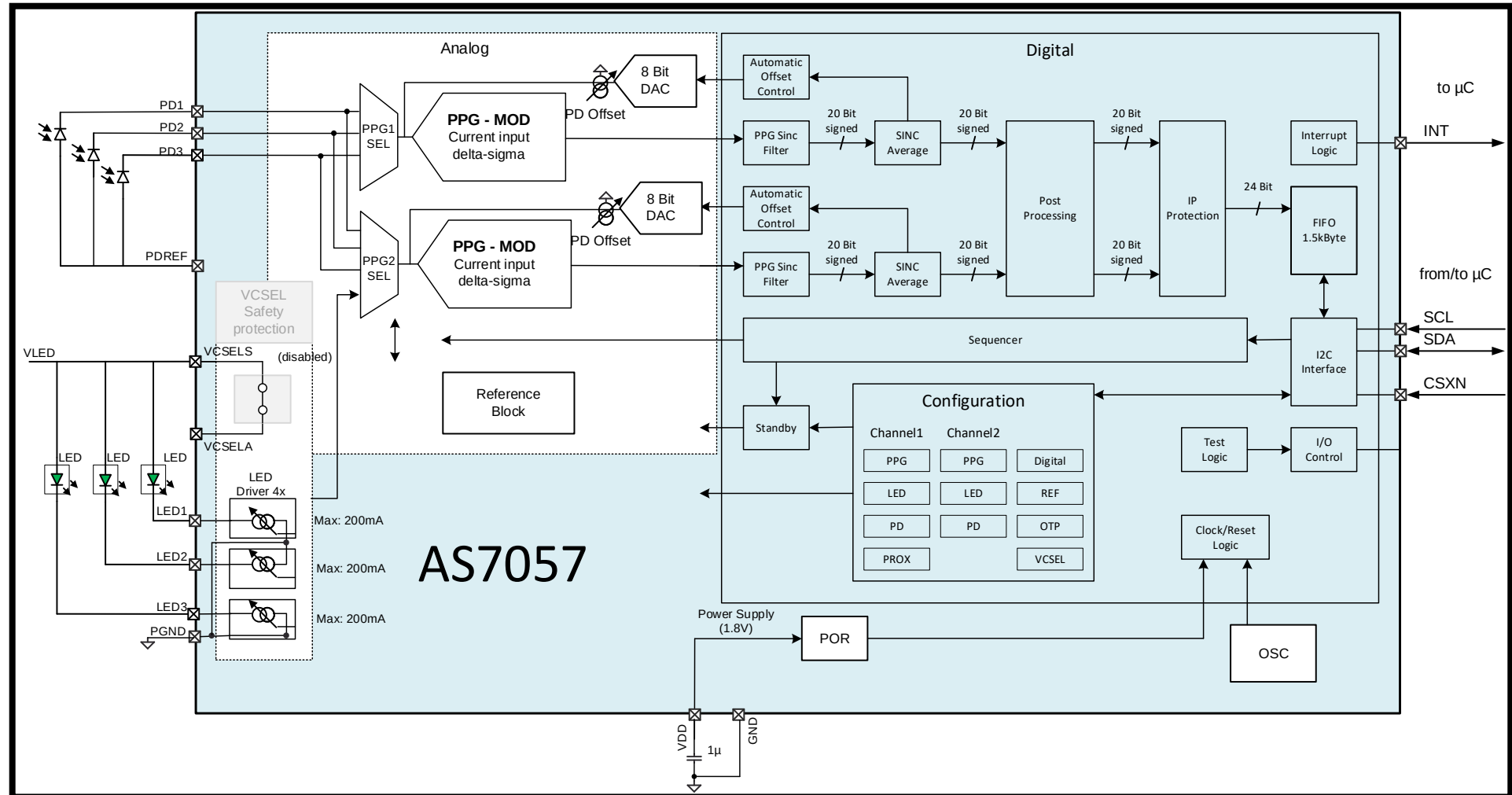


Fig. 5: AS7057 Block Diagram

AS7057 System Overview

Pin Diagram

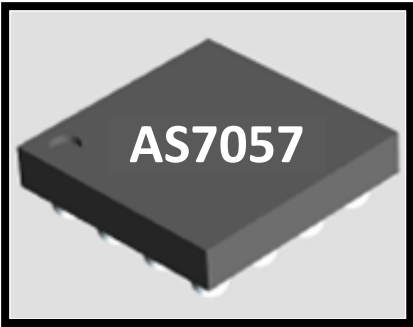


Fig 6: Top View of AS7057

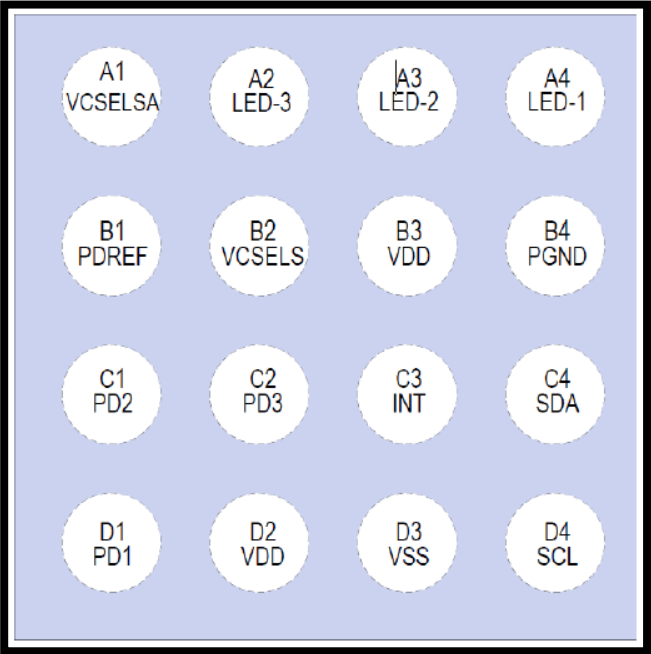
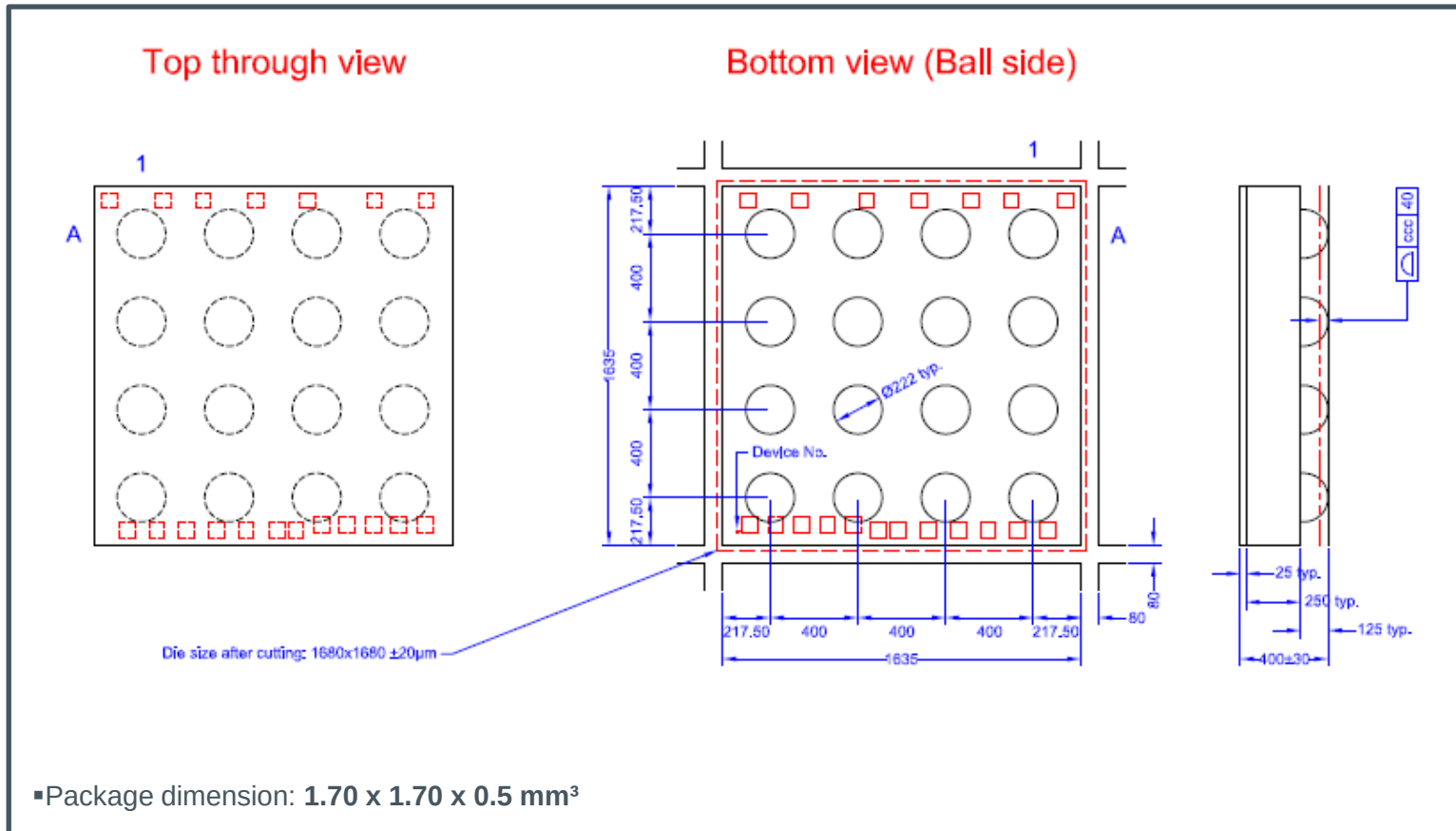


Fig. 7: AS7057 Pin Diagram

Pin Number	Pin Name	Description
A1	VCSELSA	No Connection
A2	LED3	Output LED driver 3
A3	LED2	Output LED driver 2
A4	LED1	Output LED driver 1
B1	PDREF	Reference potential for photodiodes
B2	VCSELS	Supply Pin for LED and VCSELS. This pin must always be connected to LED/VCSEL power supply
B3	VDD	Power Supply (typ. 1V8)
B4	PGND	Ground for LED drivers 1-3
C1	PD2	Photodiode input
C2	PD3	Photodiode input
C3	INT	Interrupt
C4	SDA	I ² C Data
D1	PD1	Photodiode input
D2	VDD	Power Supply (typ. 1V8)
D3	VSS	Ground
D4	SCL	Clock for I ² C

Fig. 8: Pin description

Package Dimension

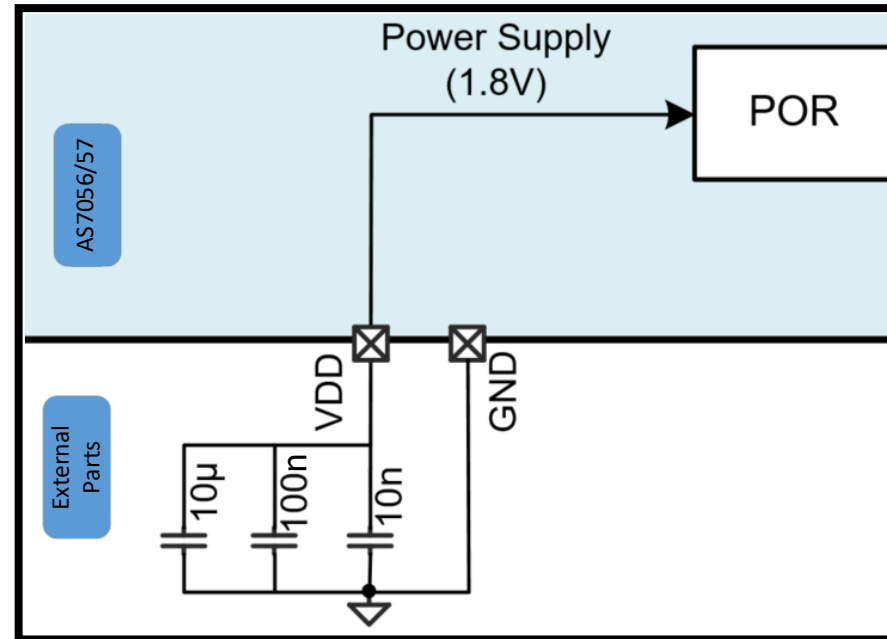


- All dimensions are in micrometers and angles are in degrees.
- Dimensioning and tolerances conform to ASME Y14.5M-1994.
- This package contains no lead (Pb).
- This drawing is subject to change without notice.

Fig. 9: AS7056/57 package Dimension

Voltage Supply Scheme

Power Supply



Symbol	Description	Min	Typ	Max	Unit
VDD	Digital power supply	1.7	1.8	1.98	V
GND	Digital Ground	0	0	0	V

Fig. 10: Voltage supply scheme

Digital IO Signals

Enable I2C interface

- A digital I2C slave interface is used for external control of the measurement setup and controls the function of all internal features. The AS7056/57 support single and burst access via I2C. Single access requires about 50% more time compared to burst access for the transmission of the same amount of data.
- The AS7056/57 includes an I2C slave using an I2C address of 0x55 fast mode (400kHz), and standard mode (100kHz) support.

Pin number	Pin Name	Description and usage
C4	SDA	I2C data, pull up 2K2 to VCC with a resistor
D4	SCL	I2C clock, pull up 2K2 to VCC with a resistor

Fig. 11: AS7056/57 I2C interface

Digital IO Signals

Interrupt PIN

- The interrupt events are processed by the interrupt manager. These interrupt events must be released for processing via an interrupt enable register.
- The interrupt status is automatically reset when the register is read (auto-zero register).

Pin number	Pin Name	Description and usage
C3	INT	Interrupt pin, pull up 10K to VCC with a resistor

Fig. 12: AS7056/57 interrupt pin description

IO Signals

Pin description – LED Drivers and Photodiodes

- The AS7056/57 provides a high amount of LED driver output and photodiode input pins. This enables the highest flexibility for several LED and photodiode arrangements in different applications.
- It can drive external LEDs and support additional photodiodes.

Pin No.	Pin Name	Description
A4	LED1/VCSEL	Output LED driver 1
A3	LED2	Output LED driver 2
A2	LED3	Output LED driver 3
D1	PD1	Photodiode input
C1	PD2	Photodiode input
C2	PD3	Photodiode input

Fig. 13: AS7056/57 Pin description – LED drivers & Photodiodes

PPG Design

LED/VCSEL Drivers

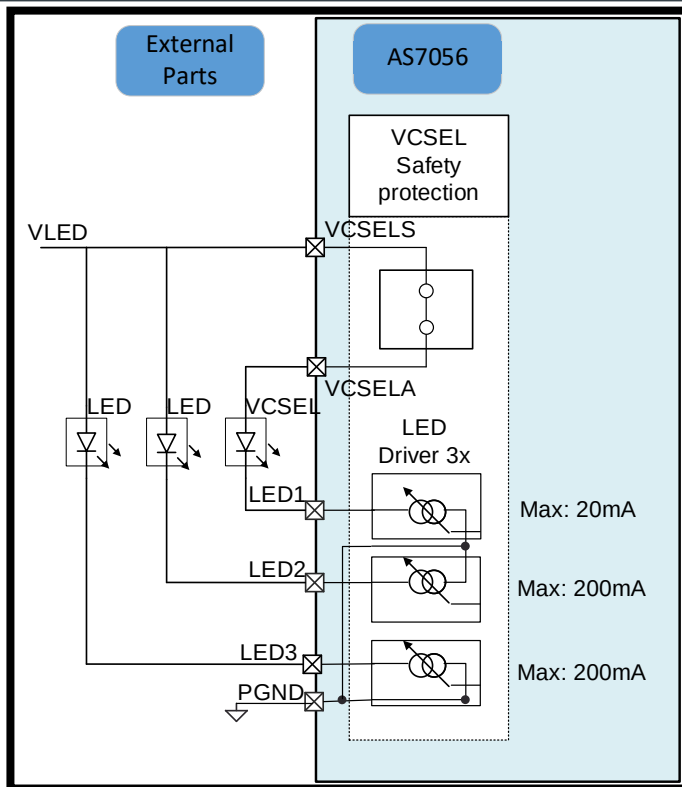


Fig. 14: AS7056 LED drivers

- 1 VCSEL driver supports up to 20 mA current
- 2 LED drivers support up to 200 mA current
- VCSEL controller (eyes safety monitor)
- Very low power consumption: down to 10 μ A @ 25 Hz sampling freq. (without LED driving current)

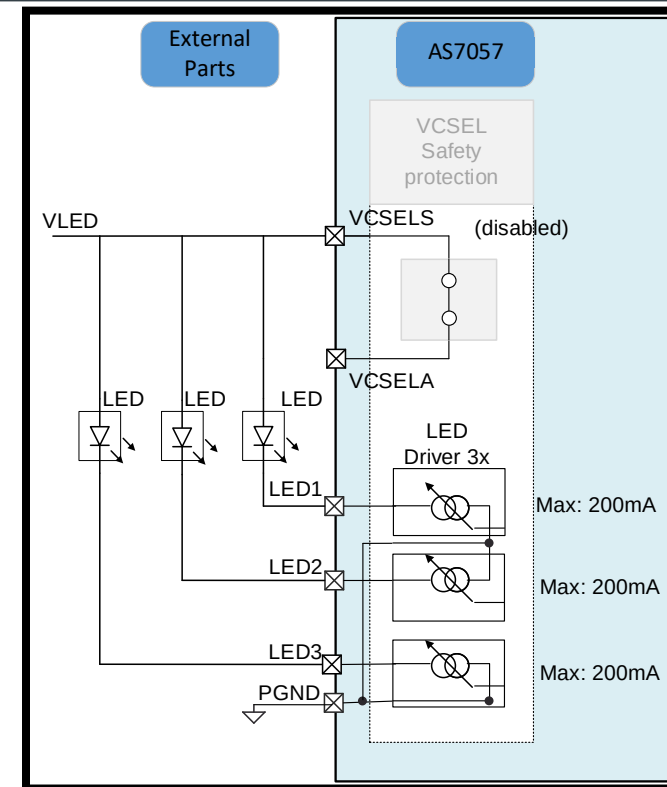


Fig. 15: AS7057 LED drivers

- 3 LEDs drivers support up to 200 mA current
- Very low power consumption: down to 10 μ A @ 25 Hz sampling freq. (without LED driving current)

PPG Design

PD inputs

- The AS7056/57 has three photodiode input pins.
- The maximum allowed photodiode capacitance connected to PPG ADC is 300pF.
- *Fig 14* shows a basic idea of the PD connections.

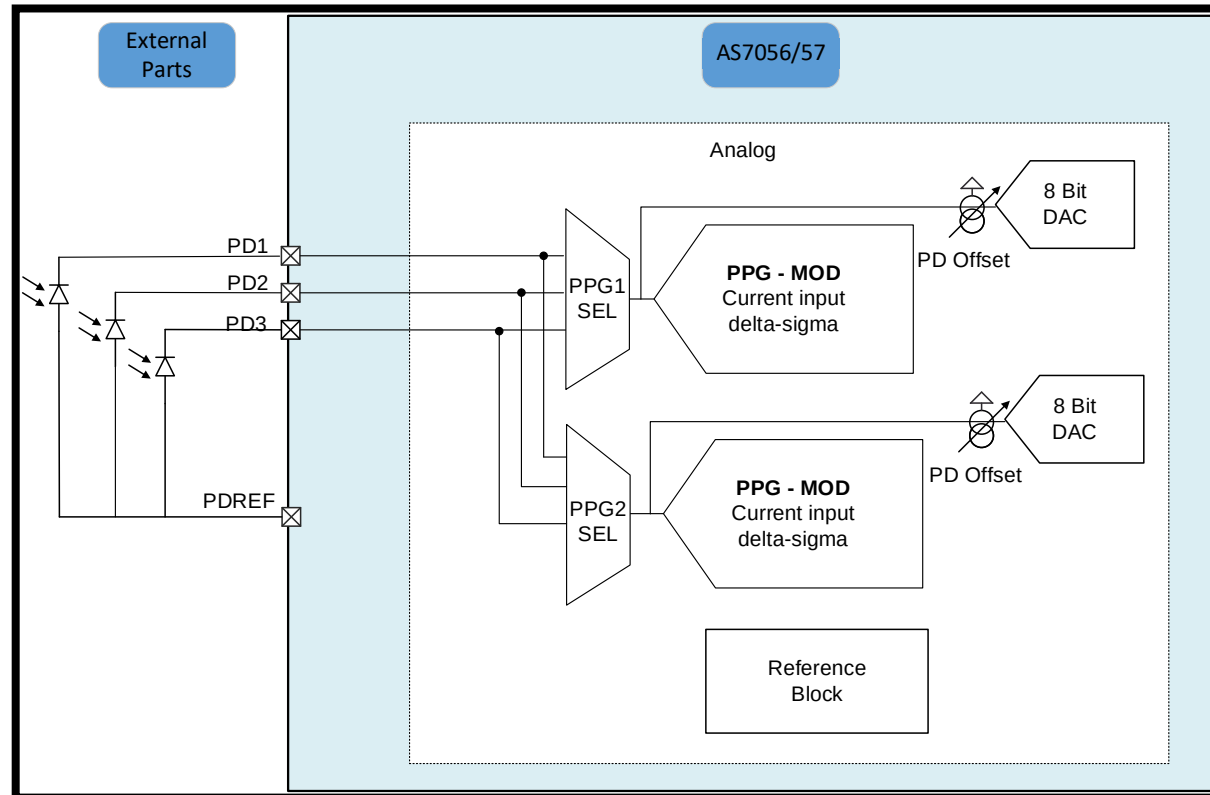


Fig. 16: AS7056/57 PPG Input

Proposal for Optical Design

Simple Design

- **ams** provides two Osram optical designs to evaluate all the functions on the AS7056/57 Biosensor and test various applications.
- It has simple and advanced designs to increase activities towards measuring HRM, SpO2, and PS.
- High-end arrangement for higher accuracy and medical grade.

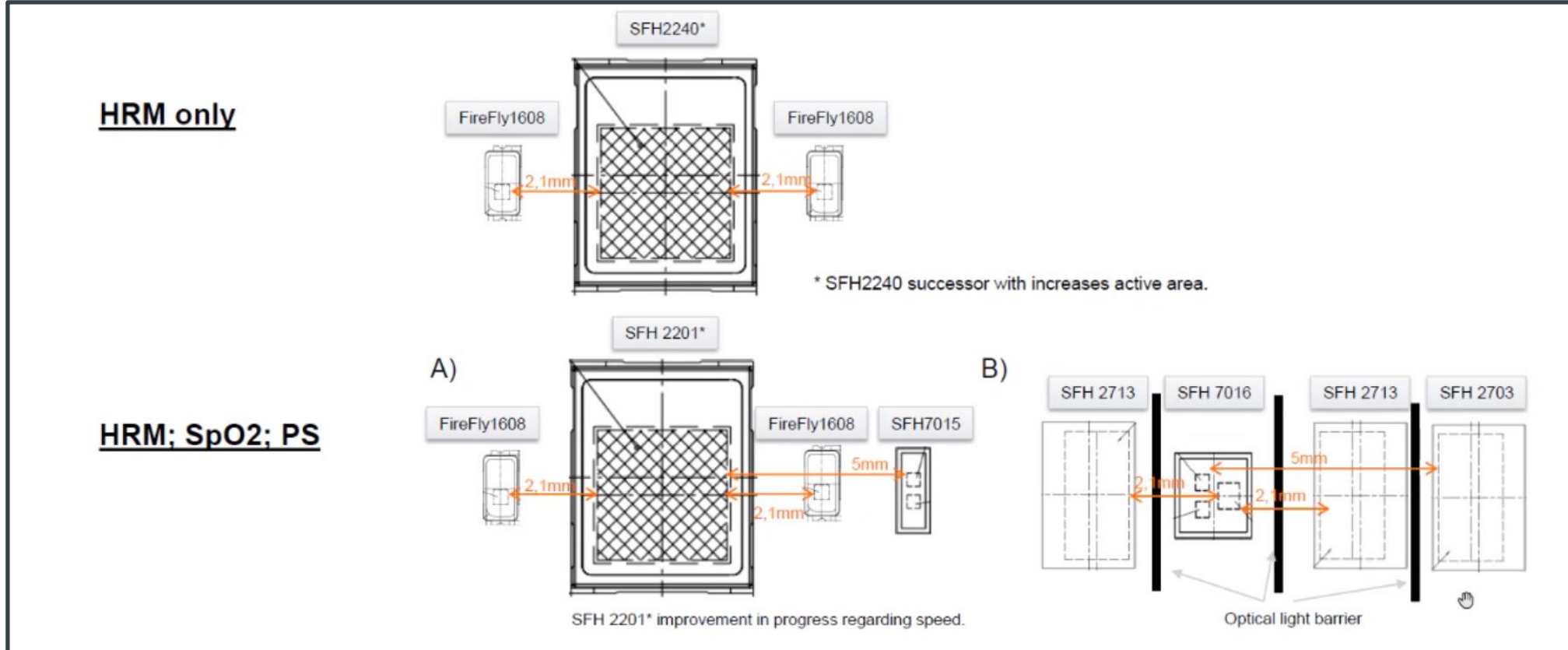


Fig. 17: Simple Design

Proposal for Optical Design

Simple Design

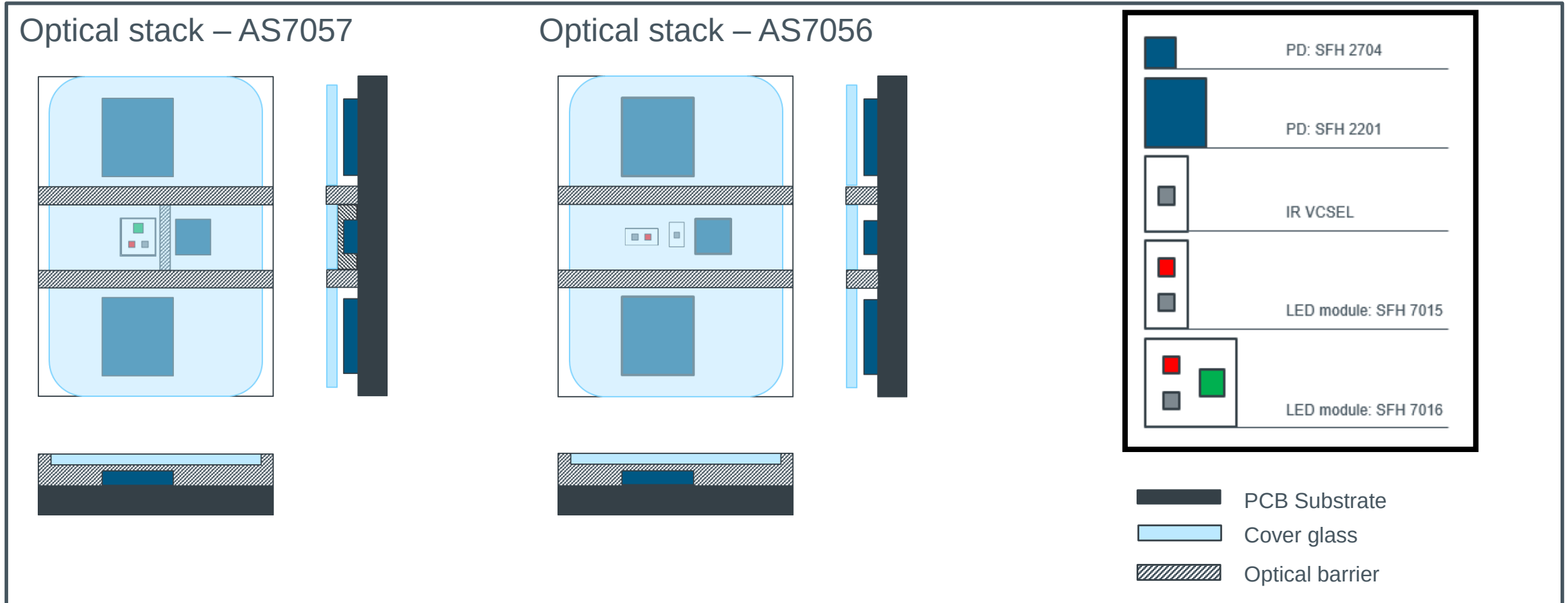


Fig. 18: Optical Design

VCSEL Requirement: It has to meet the Class1 laser safety limits including S.F.C (single fault condition) in compliance with IEC/EN 60825-1:2014.

Proposal for Optical Design

Proximity Detection

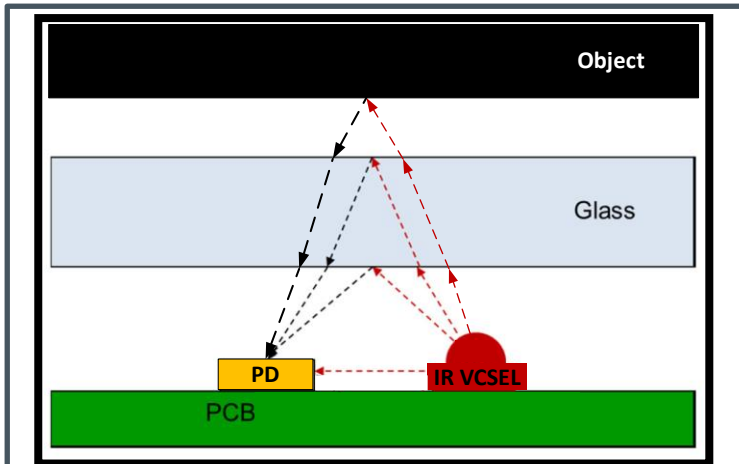


Fig. 19: Typical Proximity Detection for AS7056

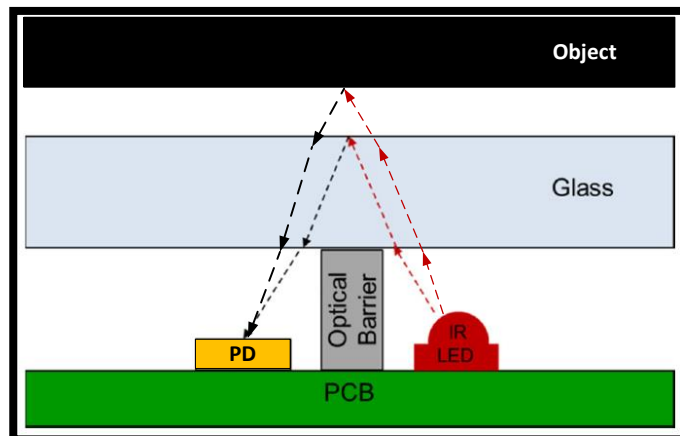


Fig. 20 : Typical Proximity Detection for AS7057

- The proximity function measures the amount of IR light received at the broadband photodiode (PD) detector.
- Large air-gap, Optical barrier, and distances between the IR-VCSEL(or IR - LED) and the broadband photodetector (PD) all are lead to increased crosstalk.
- PPG channel of AS7056/57 is used for proximity and the measurement is done with Sequencer 2.
- The user can define the ADC minimum and maximum threshold for proximity measurement. When the count is above or below the defined range, the device sends out an interrupt.
- The range must be adapted to suit the use cases.

Optical system components

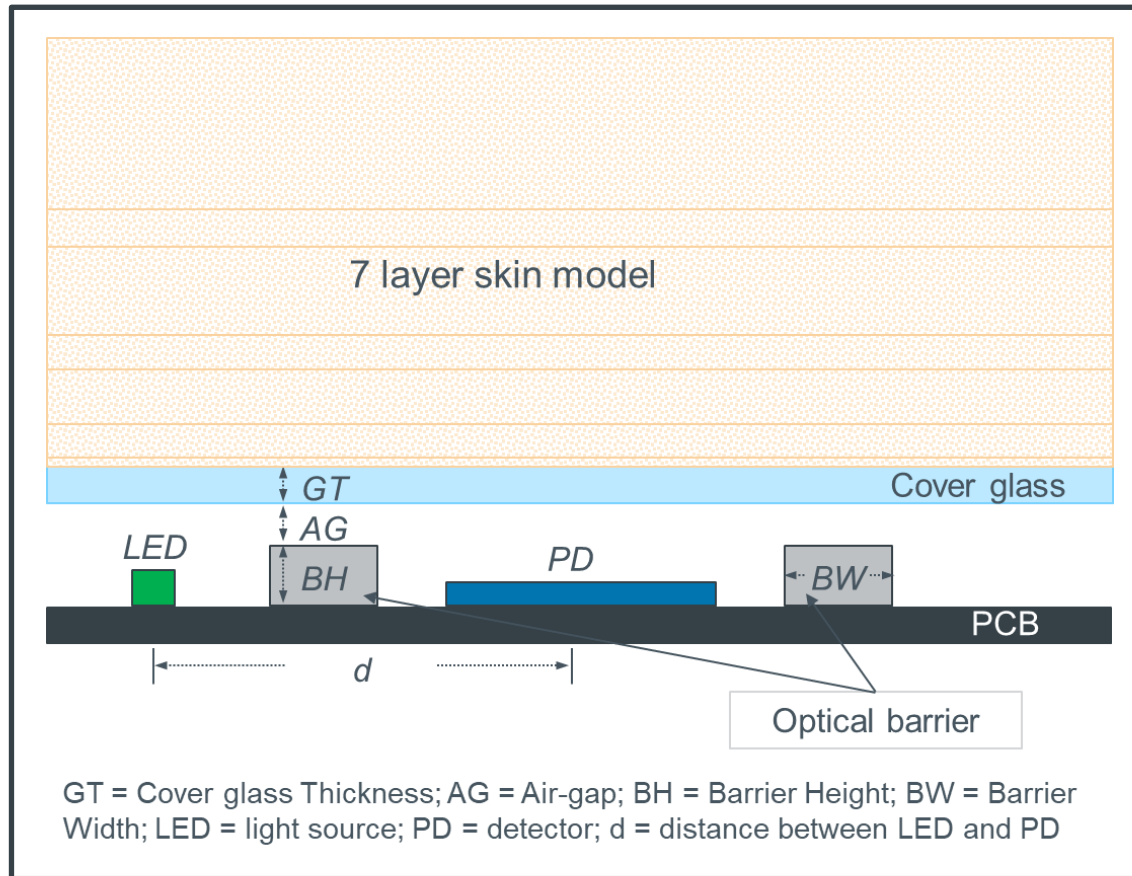


Fig. 21: Components of optical system

- The optical system consists of an AS7056/57 placed on a PCB, along with the LED and the 1 mm high optical barrier surrounding the photodiode.
- A cover glass is placed in between the source and detector and the 7-layer skin model.
- The optical properties of each optical component are provided in the table in the next slide.
- Assuming a symmetrical system, only one LED was used as a light source.
- Optical simulations were carried out using Zemax ray tracing software.
- The signal is defined as the optical power detected on the surface of the PD when the skin is in contact with the cover glass.
- Crosstalk is defined as the detected optical power without the skin. This represents the light rays reaching the detector without hitting the skin surface.
- A ratio of signal to crosstalk (SXR) is used for comparing the effect of parameters such as d , GT , AG , and BW for different LEDs.

Optical properties of the system components

*Table 1: Skin Parameters**

Skin Layer	Layer thickness (mm)	Refractive Index (generic)	skin reflectivity generic (diffused)	Absorption coefficient (1/mm)			Scattering coefficients	
				at 525 nm	at 660 nm	at 940 nm	generic	<i>g</i>
1 - Stratum corneum	0.02	1.5	0.07	0.4493	0.5488	0.7408	1.00	0.86
2 - Living epidermis	0.09	1.34	0.00	0.1353	0.4493	0.6065	1.00	0.8
3 - Papillary dermis	0.175	1.4	0.00	0.7788	0.8869	0.4966	1.00	0.9
4 - Upper blood net dermis	0.09	1.39	0.00	0.6376	0.8958	0.4966	1.00	0.95
5 - Reticular dermis	1.5	1.4	0.00	0.8607	0.8869	0.4966	1.00	0.8
6 - Deep blood net dermis	0.105	1.38	0.00	0.5488	0.9048	0.4966	1.00	0.95
7 - Subcutaneous fat	6.25	1.44	0.00	0.8607	0.9048	0.4966	1.00	0.75

* Reference for skin parameters: I.V. Meglinski and S.J. Matcher, Computer Methods and Programs in Biomedicine 70 (2):179-186, 2003.

Table 2: Optical properties of PCB and Barrier

System Component	Reflectivity		Absorption	Transmission	Remarks
	Diffused	Specular			
PCB	0.5 of 50%	0.5 of 50%	50%	0	PCB is considered to be 50% reflective.
Optical Barrier	0.5 of 20%	0.5 of 20%	80%	0	Barrier is considered to be 20% reflective.
Cover glass	As per Fresnel equations		0%		

PD-LED distance dependence of SXR

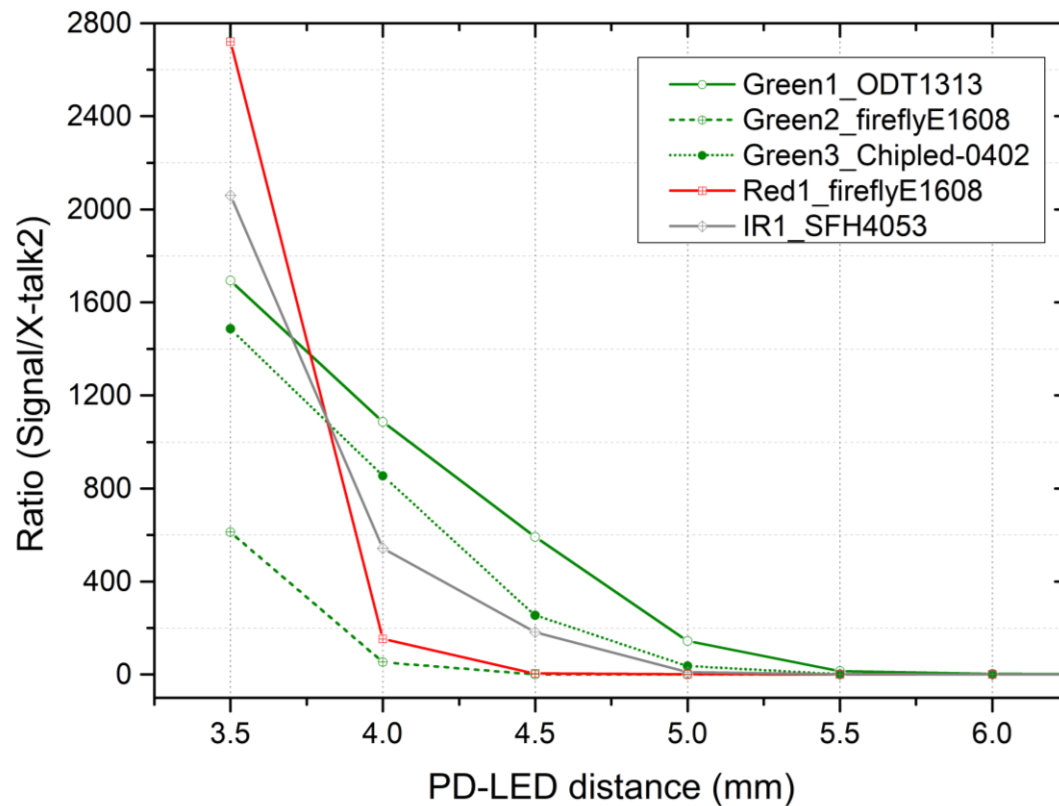


Fig. 22: Graph of SXR against PD-LED distance

- In this graph, SXR values are presented for various LEDs as a function of PD-LED distance.
- System: AG = 0.1 mm; CT = 0.3 mm; BW = 1 mm
- The minimum simulated distance is 3.5 mm.
- It is clear that the SXR values decrease rapidly with increasing PD-LED distance.

Effect of air gap on SXR

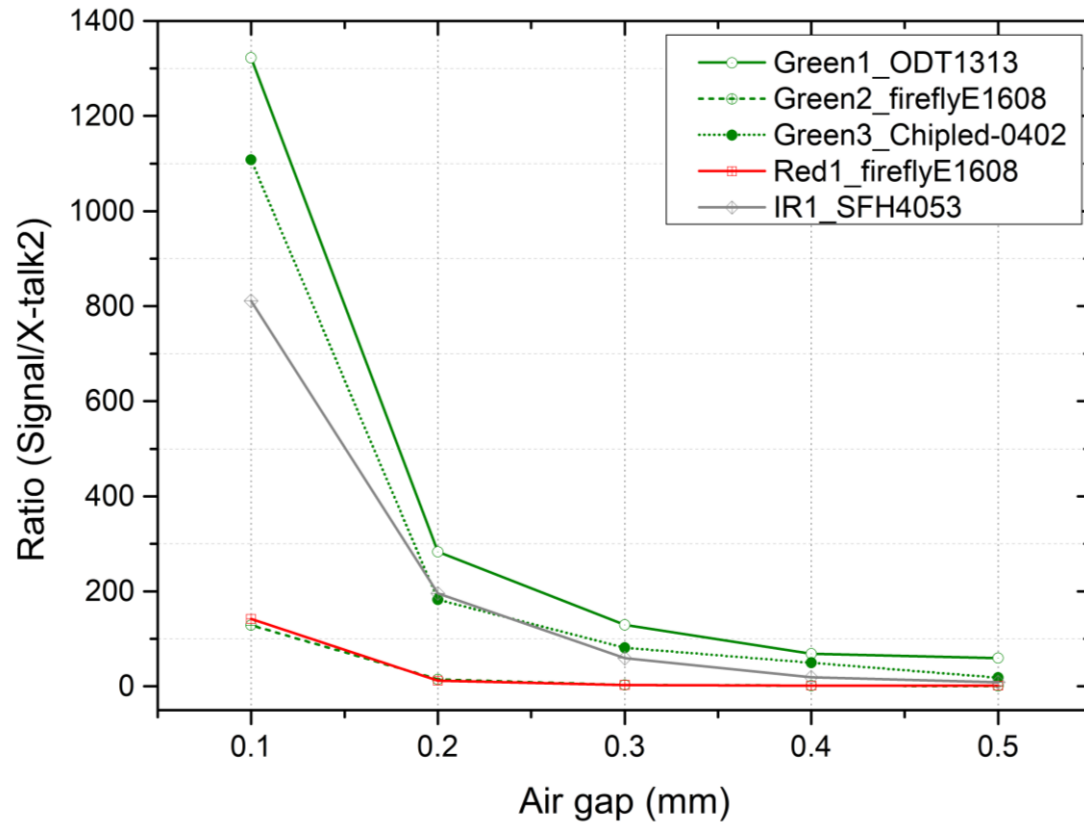


Fig. 23: Graph of SXR against Air gap

- In this graph, SXR values are presented for various LEDs as a function of air gap.
- System: $d = 0.4$ mm; $CT = 0.3$ mm; $BW = 1$ mm
- The minimum simulated air gap ($AG = 0$ mm) results in an X-talk value of 0. Hence, the SXR values for AG starting at 0.1 mm is presented.
- It is clear that the SXR values decrease rapidly with increasing AG value.

Effect of coverglass thickness on SXR

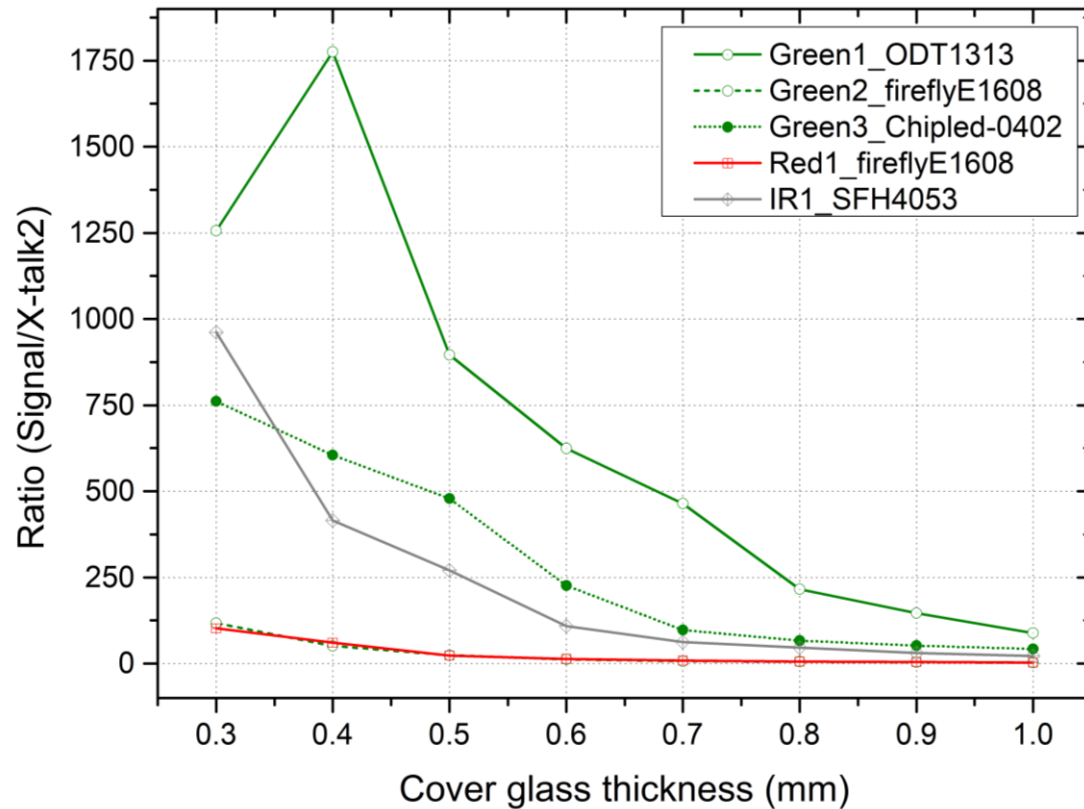


Fig. 24: Graph of SXR against the thickness of Cover glass

- In this graph, SXR values are presented for various LEDs as a function of cover glass thickness.
- System: $d = 0.4$ mm; $AG = 0.1$ mm; $BW = 1$ mm
- The minimum simulated CT is 0.3 mm.
- It is clear that the SXR values decrease with increasing CT, except for the green LED chip (ODT1313UX3.A3).
- The reason for this exceptional behavior requires further investigation.
- It can also be seen from the graph that the LEDs in similar packages result in similar SXR values.

Effect of barrier width on SXR

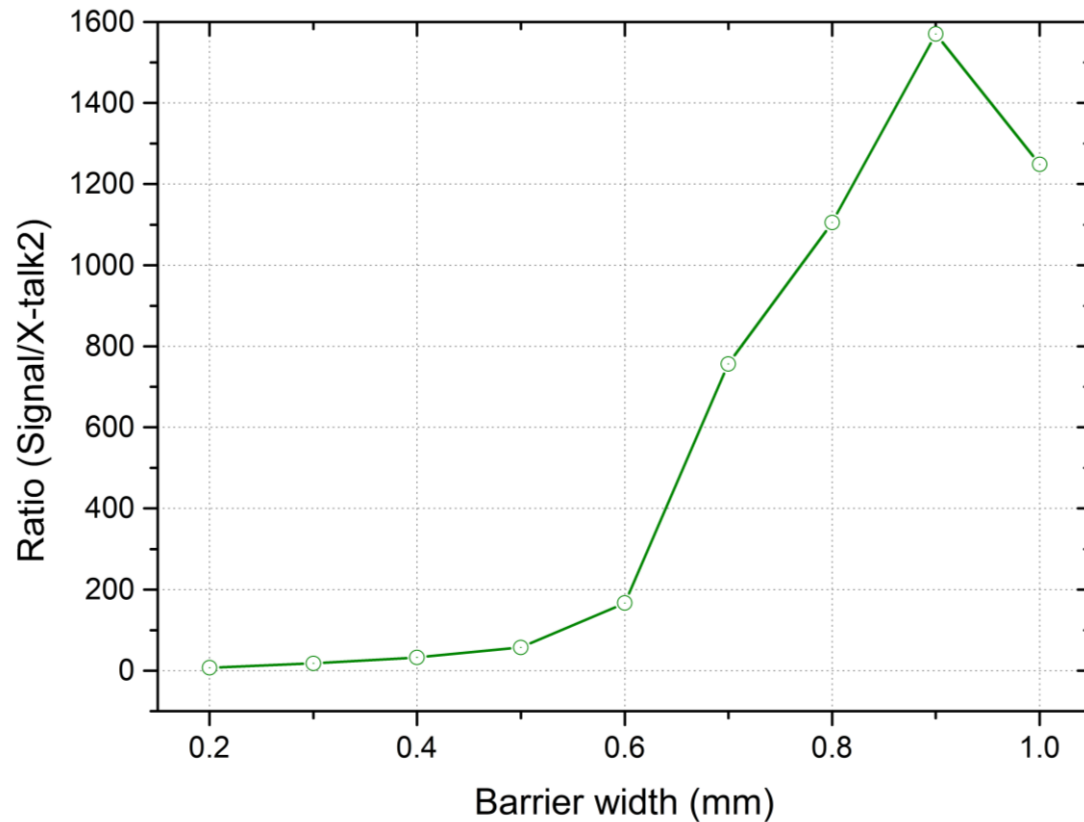


Fig. 25: Barrier width dependence of SXR

- In this graph, SXR values are presented for the green LED chip as a function of optical barrier width.
- System: $d = 0.4$ mm; $AG = 0.1$ mm; $CT = 0.3$ mm
- The minimum simulated BW is 0.2 mm.
- The SXR values increase with increasing optical barrier width until $BW = 0.9$ mm. A further increase in BW results in a decrease in the SXR value.

Effect of barrier type on optical power

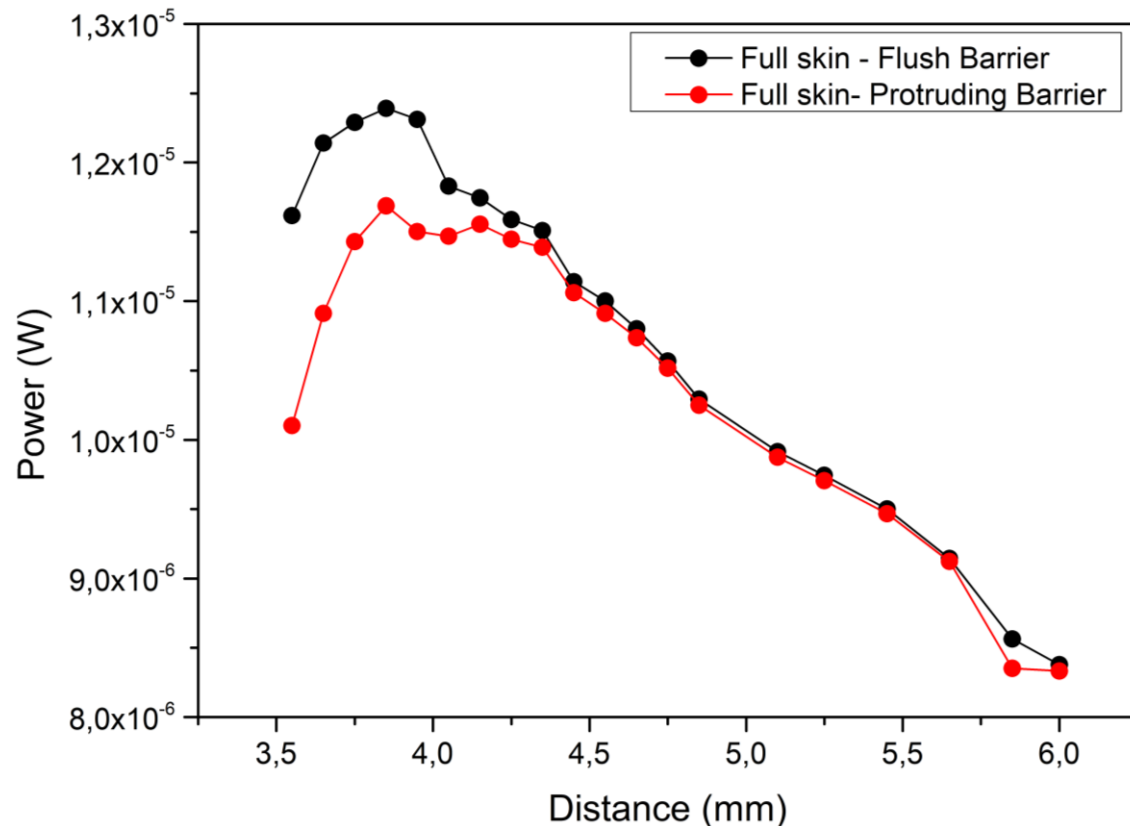
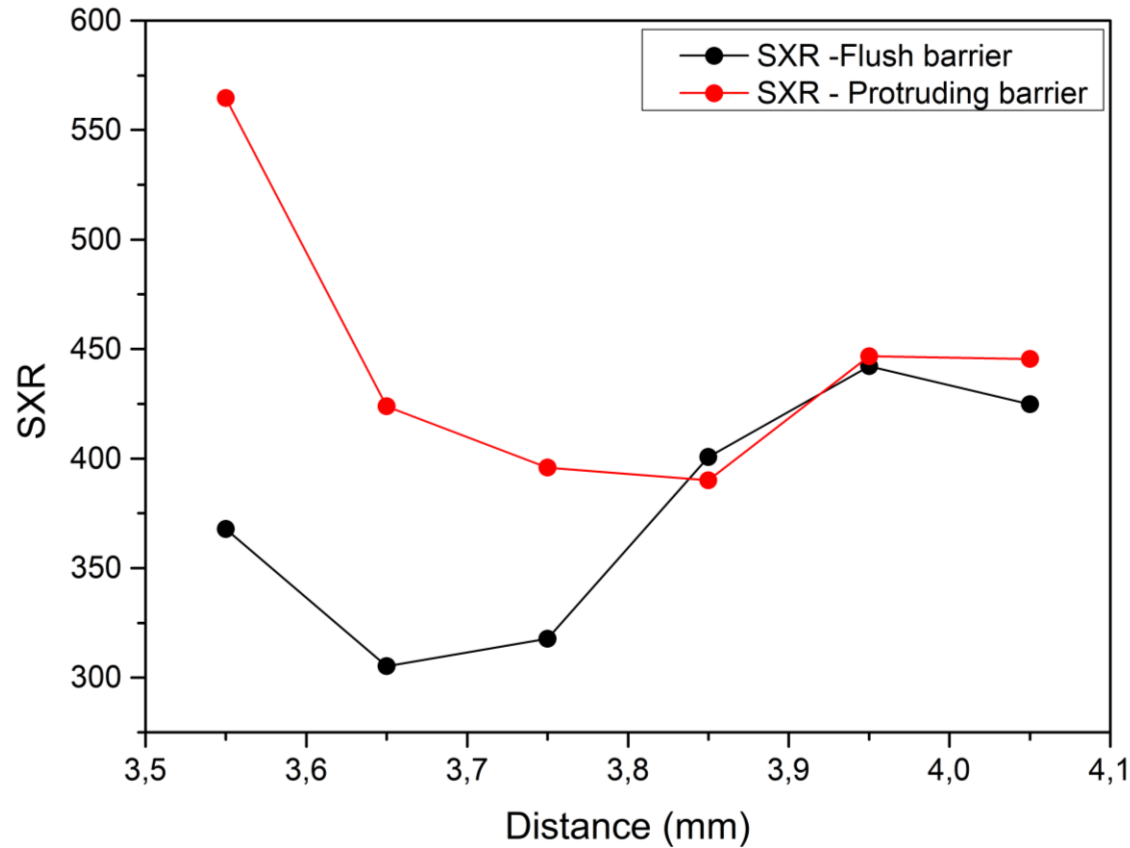


Fig. 26: Graph of Power against Distance

Optical power received on the photodiode at various LED-PD separation for the two types of optical barriers. LED: green (CT DELSS1.12), cover glass thickness: 0.3 mm, Air gap: 0 mm.

- Two sets of simulations were done to understand the effect of the optical barrier type on the optical power received on the detector.
- Barrier types:
 - Flushed barrier – This barrier type ends just below the cover glass.
 - Protruding barrier – This barrier type separates the cover glass into multiple segments.
- Thus, by design, the protruding barrier is taller than the flushed barrier.
- It is seen from the simulation results that for smaller PD-LED separation, the optical power received on the photodiode is higher with a flushed type barrier. For distances larger than 4 mm, the difference in power reduces gradually.
- The crosstalk on the other hand is expected to be higher with the flushed barrier type.

Effect of barrier type on optical power



The plot of SXR values for flushed and protruding barrier shows that at smaller LED-PD separation distances, the protruding barrier would result in better SXR values.

Fig. 27: Graph of SXR against Distance



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